SEMESTER 2 EXAMINATIONS 2021/2022

MODULE: CA4023 - Natural Language Technologies

PROGRAMME(S):
DS BSc in Data Science
CSPT PhD-track

YEAR OF STUDY: 1,4

EXAMINER(S):
Jennifer Foster (Internal) (Ext:5263)

TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer Question 5 and 3 other questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 25]***

Q 1(a) **[5 Marks]**

Explain, using examples, what is meant by the term **Document Classification**.

Q 1(b) **[5 Marks]**

What is the difference between **Logistic Regression** and **Multinomial Logistic Regression**? What is the purpose of the following function in Logistic Regression?

$$\frac{1}{1 + e^{-x}}$$

Why is the above function not used in Multinomial Logistic Regression and what function is used instead?

Q 1(c) **[5 Marks]**

Explain the role of a **loss function** when training a Logistic Regression classifier.

Q 1(d) **[10 Marks]**

Given the following training corpus

Document	**Class**
Not great value for money	negative
Keep your money	negative
Good value for money	positive
Very reliable	positive

what label would a Naïve Bayes sentiment classifier which uses the words in a document as features assign to the following "document"?

`Very good value for money`

Show the steps which the classifier follows to arrive at a prediction.

[End of Question 1]

QUESTION 2 ***[TOTAL MARKS: 25]***

Q 2(a) **[10 Marks]**

Given the following training corpus

```
<s> a b </s>
<s> b a </s>
<s> b c </s>
<s> c a </s>
```

what probabilities will an unsmoothed bigram language model assign to the following sentences?

```
<s> a </s>
<s> b </s>
<s> b b </s>
<s> b a </s>
<s> c a b </s>
```

Show how you arrived at your answer.

Q 2(b) **[5 Marks]**

Explain, using an example, what is meant by **language model smoothing** and briefly explain one smoothing technique.

Q 2(c) **[4 Marks]**

State one difference and one similarity between a language model trained using a **recurrent neural network** and one trained using a **transformer network**.

Q 2(d) **[6 Marks]**

Language models trained using a neural network are typically more accurate than n-gram language models. Explain, using an example, why this is the case.

[End of Question 2]

QUESTION 3 ***[TOTAL MARKS: 25]***

Q 3(a) **[5 Marks]**

Explain, using examples, what is meant by the term **Sequence Labelling**.

Q 3(b) **[10 Marks]**

State the emission (*word|tag*) probabilities that result from training a Hidden Markov Model (HMM) part-of-speech tagger using the following training corpus:

a D green A chair N
the D green N party N meeting N
chair V the D meeting N
they N party V

Assume that the label set contains the following tags: *N (noun), A (adjective), V (verb), D (determiner)*
Name and briefly explain the other probability distribution modelled by an HMM.

Q 3(c) **[3 Marks]**

Name a discriminative alternative to HMMs for sequence labelling, and briefly explain its advantage over HMMs.

Q 3(d) **[7 Marks]**

Recurrent neural networks are often used for sequence labelling. Sketch the architecture of an RNN and show how it is used to label a sequence.

[End of Question 3]

QUESTION 4 **[TOTAL MARKS: 25]**

Q 4(a) **[7 Marks]**

Consider the following term-document matrix:

	The Gruffalo	**Cinderella**	**Sleeping Beauty**	**Snow White**	**The Snow Queen**
mouse	**20**	**5**	**0**	**0**	**0**
gruffalo	**25**	**0**	**0**	**0**	**0**
princess	**0**	**20**	**22**	**25**	**25**
pumpkin	**0**	**6**	**0**	**0**	**0**
snow	**0**	**0**	**2**	**50**	**40**

Define what is meant by TF-IDF using examples from the above term-document matrix to illustrate your answer.

Note: you do not have to work out the exact TF-IDF values.

Q 4(b) **[4 Marks]**

Explain, using an example, how a word co-occurrence matrix differs from a term-document matrix.

Q 4(c) **[7 Marks]**

The **Skipgram Negative Sampling word2vec** classifier uses logistic regression to predict whether two words co-occur. Explain, using an example, how training data is obtained for this classifier.

Q 4(d) **[7 Marks]**

Explain, using an example, the difference between **static** and **dynamic** neural word vectors, and provide examples of models that can be used to learn each type.

[End of Question 4]

QUESTION 5 ***[TOTAL MARKS: 25]***

Q 5(a) **[12 Marks]**

The following have been identified as potential harms associated with large neural language models such as GPT and BERT:

1. Environmental costs
2. Amplifying harmful stereotypes
3. Exclusion of social groups
4. Exclusion of languages
5. Propaganda and fake news generation
6. Raised expectations regarding the extent to which such models understand language

For each one, briefly explain the potential threat.

Q 5(b) **[6 Marks]**

For each potential threat listed above, suggest a way that such a threat can be mitigated.

Q 5(c) **[7 Marks]**

What are the potential benefits of large neural language models? In your opinion, do the benefits outweigh the risks?

[End of Question 5]

[END OF EXAM]